CHECK WHAT YOU LEARNT

Check Yourself - Unit 2

A. Choose the correct answer

1. The actual computer on an Arduino Uno is
   (a) the USB socket (b) the ATmega328P chip (c) the blue board (d) the crystal
2. The 16 MHz crystal provides
   (a) power (b) the clock (c) memory (d) the reset
3. Your compiled program is stored in
   (a) SRAM (b) flash memory (c) EEPROM (d) the ADC
4. The ATmega's ADC is 10-bit, so it reports
   (a) 10 values (b) 256 values (c) 1024 values (d) 1023 volts
5. One ADC step across a 5V range is about
   (a) 4.9 millivolts (b) 49 millivolts (c) 0.5 volts (d) 5 volts
6. The bootloader's job is to
   (a) run your program (b) receive your code and write it to flash (c) measure voltage (d) power the board

B. Fill in the blanks

1 Variables live in _______________ while the program is running.

2 A 10-bit ADC reports numbers from ______ to ______.

3 The smallest change a system can detect is called its _______________.

4 Ready-made code written by somebody else is called a _______________.

C. Answer in one or two lines

1 Describe the four stages between pressing Upload and your program running.

2 A pin sits at 3.75 volts. Roughly what will analogRead() return? Show your working.

3 Explain why a program with many long text messages may run out of memory.

4 Why is using a library not cheating?

UNIT 3

Basic Project: Fire Security Alarm System

Sessions 6, 7 and 8

Your first machine has a serious job. It must notice a fire quickly, be certain enough not to cry wolf, and once it has decided, refuse to go quiet until a human says so. Every one of those requirements changes the code.

By the end of this unit you will be able to:

- Explain how a flame sensor detects fire without measuring heat.
- Combine two sensors so that both must agree before an alarm sounds.
- Implement a latching alarm with a manual reset.
- Test honestly for false positives and false negatives.

SESSION 6

one class